

GUIDELINES FOR ASSESSING BIOINFORMATICS PROJECT REPORT

The following are the generic marking criteria upon which pieces of work are assessed. Assessment criteria are broken down into the dimensions: Summary (S), Knowledge (K), Cognitive Skill (C) and Presentation (P). See the glossaries below for further information in the dimensions and assessment criteria.

Assessment criteria	Summary statement (S)	Knowledge (K)	Cognitive Skill (C) Lower order cognitive skills	Presentation (P)
0-100 point scale		1. Depth and breadth of appropriate knowledge. 2. Evidence of independent reading related to that expected from supervisor's advice. 3. Integration of material. 4. Presence of irrelevant and tangential material. 5. Presence of factual errors. 6. Presence of gaps or omissions.	1. Understanding of underlying concepts. Skills that demonstrate critical ability: 2. Apply: appropriate data analysis methods. 3. Analysis: Full analysis of data. 4. Evaluation: Interpretation of results and judgements made with regards to methodology, hypothesis etc. 5. Create/Synthesis.	1. Structure and narrative. 2. Clarity of scientific writing. 3. Quality of figures, tables and diagrams (if appropriate). 4. Consistency of style.
100 94 83 (Outstanding)	Intended learning outcomes satisfied to an outstanding standard with no areas of weakness. Work would be worthy of dissemination under appropriate conditions.	• Outstanding depth and breadth of knowledge. • Exceptional evidence of relevant independent reading beyond that expected from supervisor's advice. • Exceptional integration of material. • All material is relevant. • Answer is free from factual errors. • No gaps or omissions.	• Evidence that the underlying concepts are completely mastered. • Displays an outstanding command of critical ability. • Outstanding evidence of originality and synthesis.	All presentation criteria satisfied to an outstanding standard.
78 75 72 (Excellent)	Intended learning outcomes satisfied to an excellent standard with very few areas of weakness and no significant errors.	• Excellent depth and breadth knowledge. • Excellent evidence of relevant independent reading beyond that expected from supervisor's advice. • Substantial integration of material. • All material is relevant. • Answer includes only minor factual errors if any. • No gaps or omissions.	• Evidence that the underlying concepts are fully understood. • Displays an excellent command of critical ability. • Evidence of originality and excellent synthesis.	All presentation criteria satisfied to an excellent standard.

68 65 62 <i>(Good to Very Good)</i>	Intended learning outcomes satisfied to a good or very good standard with few areas of weakness or significant errors.	• Good to very good depth of knowledge. • Good evidence of relevant independent reading. • Good integration of material. • May contain limited irrelevant or tangential material. • Largely free from factual errors.	• Evidence underlying concepts are understood. • Displays a good to very good command of critical ability. • Good evidence of synthesis.	All presentation criteria satisfied to a good or very good standard.
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		• May contain some minor gaps or omissions.		
58 55 52 <i>(Adequate)</i>	Intended learning outcomes satisfied with some errors or omissions, but no fundamental errors or omissions.	• Adequate depth and breadth of knowledge. • Adequate evidence of relevant independent reading but most references included are those expected from the direction outlined by the supervisor. • Some attempt to integrate material. • May contain some irrelevant or tangential material. • May contain some factual errors. • May contain some gaps or omissions.	• Evidence underlying concepts are largely understood. • Adequate evidence of critical ability • Adequate evidence of synthesis	All presentation criteria satisfied to an adequate standard.
48 45 42 <i>(Poor)</i>	Most intended learning outcomes satisfied but with some fundamental errors or omissions.	• Little depth and breadth of knowledge. • Poor evidence of independent reading and references are only those expected from the directions outlined by the supervisor. • Little attempt to integrate material. • May contain substantial irrelevant or tangential material. • Contains significant factual errors. • Gaps and/or omissions.	• Evidence most underlying concepts have been largely understood. • Little evidence of critical ability. • Little evidence of synthesis	Most presentation criteria satisfied but to a poor standard.

35 (<i>Very Poor</i>)	Some intended learning outcomes satisfied with fundamental errors or omissions. Constitutes a borderline fail.	• Very little depth and breadth of knowledge. • No evidence of relevant independent reading. • No attempt at integration of material. • May contain substantial irrelevant or tangential material. • Contains serious errors. • Large gaps and/or significant omissions.	• Insecure understanding of underlying concepts. • No evidence of critical ability. • No evidence of synthesis	Some presentation criteria satisfied, but to a poor standard.
29 22 15 7 (<i>Inadequate</i>)	Most Intended learning outcomes clearly NOT satisfied, with multiple fundamental errors or omissions.	• Inadequate depth and breadth of knowledge. • No evidence of relevant independent reading. • No attempt at integration of material. • May contain substantial irrelevant or tangential material. • Contains catastrophic errors. • Large gaps and/or significant omissions.	• Deficient grasp of underlying concepts. • No evidence of critical ability. • No evidence of synthesis.	Presentation criteria not satisfied.
0 (<i>absent</i>)	S: Fails to provide assessable answer to the question, absent answer, or no evidence that any intended learning outcomes have been met.			

GLOSSARY OF ASSESSMENT CRITERIA

*Assessment criteria in bold represent detrimental aspects of an answer. A good answer will not have these attributes.

Descriptor Theme	Assessment Criteria	Description
Summary (S)	The summary statement explains how well the intended learning outcomes have been met and the presence of any errors or omissions which describes how completely the question has been answered.	
Knowledge (K) – Subject matter content.	Depth and breadth of knowledge	'Depth of knowledge' describes the level of detail in an answer. 'Breadth of knowledge' describes the range and scope of material covered in an answer. There will be essential, expected and desirable topics from the taught material to include in an answer. The depth and breadth of knowledge can be further described through the presence of factual knowledge, conceptual knowledge, and procedural knowledge (see below).
	Factual Knowledge	The use of accurate terminology and specific detail. This level of knowledge is typically associated with the Blooms skill 'remember'.
	Conceptual Knowledge	The use of classification, categories, described principles and ability to generalize. This level of knowledge is typically associated with the Blooms skill 'Understand'.
	Procedural Knowledge	Use of subject-specific skills, techniques, methods and criteria. This level of knowledge is typically associated with the Blooms skill 'Apply'.
	Evidence of independent reading	Evidence of study beyond the bounds of what has been taught.

	Integration of material across the programme.	Including material in your answer from multiple units across your programme of study.
	Presence of irrelevant and tangential material*	Material that is not relevant to answer the question or address the title.
	Presence of factual errors*	Inaccuracies in factual information.
	Presence of gaps or omissions*	There are key topics missing in the answer that were needed to fully answer the question or address the title.
Cognitive Skill (C) –what is to be done with or to the content.	Remember	Retrieval of relevant knowledge from memory (NA for open book assessments).
	Understand	Understanding of underlying concepts through interpreting, exemplifying, classifying, summarizing, inferring, comparing, or explaining information.
	Apply	Carrying out or using a procedure in a familiar or unfamiliar task.
	Analyse	Breaking material into its constituent parts and detecting how the parts relate to one another and to an overall purpose through discriminating, selecting, focusing, finding coherence, integrating, outlining, structuring, or deconstructing information.
	Evaluate	Making judgements, coordinating, detecting, monitoring and critiquing information.
	Create/Synthesis	Putting parts or information together to form a novel, coherent whole or to make a new product or create a new meaning. Evidence of <u>originality</u> beyond the bounds of what has been taught. Other verbs: Generating, hypothesising, planning, designing, producing, and constructing.
Presentation, structure and narrative. (P) – quality of structure, written English and scientific accuracy.	Structure	Clear and logical organisation of the writing (for example headings may add clarity to the structure).
	Narrative	The clear and logical organisation of arguments
	Clarity of scientific writing	The use of scientific language and technical vocabulary rather than colloquial language. Significantly poor grammar, or similar, reduces the clarity of writing.
	Quality of figures, tables and diagrams.	The inclusion of appropriate figures and diagrams, with a clear, concise and informative legend in the correct position.
	Consistency of style	Accurate and consistent formatting of citations and references, text and format.

GLOSSARY AND EXAMPLES FOR THE COGNITIVE SKILLS (C) DIMENSION

Categorizing the verb phrases and examples of related assessment formats for each verb. A “constructed response” task refers to students giving open response answers. A “selected response” task refers to Multiple Choice Questions.

Bloom's Verb	General description	Verb subcategories	Alternative names	Definition of subcategories	Examples of assessment formats
Remembering	Retrieving relevant knowledge from longterm memory	Recognizing	Identifying	Locating knowledge in long-term memory that is consistent with presented material	- Verification (e.g. choose if information is current or incorrect). - Matching (e.g. how each item in one list corresponds to an item in another list). - Forced choice (e.g. multiple choice).

		Recalling	Retrieving	Retrieving relevant knowledge from long term memory	- Low cueing (e.g. What is a meter?). - High cueing (e.g. In the metric system, a meter is a measure of ___?). - Embedding – a recall task included with the context of a large problem (e.g. recall the formula for the area of a circle to solve a word problem that requires that formula).
Understanding	Determining the meaning of instructional messages.	Interpreting	Clarifying Paraphrasing Representing Translating	Changing from one form of representation to another e.g. Paraphrasing	- Constructed response - supply an answer. - Selected response - choose an answer. - Information is presented in one form and required to construct or select the same information in a different form. Information given in the task must be new (not encountered during instruction).
		Exemplifying	Illustrating Instantiating	Finding an example or illustration of a concept or principle	- Constructed response - create an example. - Selected response - select an example from a given set (e.g. 'which of these is an inorganic compound (a) iron, (b) protein, (c) blood, (d) leaf mould').
		Classifying	Categorizing	Determining that something belongs to a category e.g. classify observed or described cases of X	- Constructed response - Student given an instance and must produce its related concept or principle. - Selected response – given an instance and select concept from a list. - Sorting task - Given a set of instances and must determine which ones belong to a category and which does not.

		Summarizing	Abstracting Generalizing	Abstracting a general theme or major points (e.g. write a short summary of X)	- Constructed response task - Read a passage and write an appropriate title. - Selected response task - Read a passage and select the most appropriate title from a list of four possible titles or rank them in order of best 'fit' to the passage.
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		Inferring	Concluding Extrapolating Interpolating Predicting	Draw a logical conclusion from presented information	- Completion task - Given a series of items and determine what will come next. - Analogy task - Analogy of the form A is to B as C is to D, and student fills in a blank. - Oddity task - given three or more items and must determine which does not belong.
		Comparing	Contrasting Mapping Matching	Detecting correspondences between two ideas (e.g. compare X with X)	Mapping – show how each part of one object, idea, problem or situation corresponds to each part of another object, idea or problem.
		Explaining	Constructing models	Construct a cause-and-effect model of a system	- Reasoning – offer a reason for a given event. - Troubleshooting – asked to diagnose what might have gone wrong in an example situation - Redesigning – change the system to accomplish some goal. - Prediction – how a change in one part of the system will effect a change in another part of the system.
Applying	Carrying out or using a procedure in a given situation	Executing	Carrying out	Applying a procedure to a familiar task	Given a familiar task that can be performed using a well-known procedure.
		Implementing	Using	Applying a procedure to an unfamiliar task	Given an unfamiliar problem that must be solved. Must demonstrate the procedure needed to solve the problem and/or solve the problem using the procedure.
Analyzing	Breaking material into its constituent parts and detecting how the parts relate to one another and to an overall structure or purpose	Differentiating	Discriminating, focusing, selecting	Distinguish between relevant and irrelevant material	Given some material and asked to indicate which parts are most important or relevant.
		Organizing	Finding coherence Integrating Outlining Structuring	Determine how elements fit or function within a structure	Imposing a structure on material e.g., an outline, table, matrix or hierarchical diagram. E.g., produce a written outline of a passage, selected which four of alternative graphical hierarchies best correspond to the organization of a passage.
		Attributing	Deconstructing	Determine points of view, bias, values or intent in presented material	Present material and asking student to construct or select a description of the point of view, intentions etc. of the material.
Evaluating	Making judgements based on	Checking	Coordinating Detecting Monitoring	Detecting inconsistencies within a process, detecting	Carrying out a solution to a problem or performing a task, where one is concerned with the consistency of the actual implementation.
	criteria and standards			effectiveness of a process	

		Critiquing	Judging	Detecting appropriateness or effectiveness of a procedure e.g. judge which of two methods is the best way to solve a problem	Asked to critique a hypothesis or creation (their own or by someone else).
Creating	Putting elements together to form a novel, coherent whole or make an original product.	Generating	Hypothesizing	Coming up with alternative hypothesis based on criteria.	Produce alternatives or hypotheses Consequences task; list the possible consequences of a certain event; Uses tasks; list all the possible uses for an object.
		Planning	Designing	Design a procedure to accomplish a task (e.g. plan a research paper on X)	Develop worked-out solutions, describe solution plans or select solution plans for a given problem.
		Producing	Constructing	Inventing a product	Design tasks, create a product that corresponds to a certain specification.

GLOSSARY AND EXAMPLES FOR THE KNOWLEDGE (K) DIMENSION

Categorizing the noun phrases (i.e. 'Knowledge').

Bloom's Verb	Description	Verb sub-categories Knowledge of....	Examples
Factual Knowledge	The basic elements that students must know to be acquainted with a discipline or solve problems in it.	Terminology	Technical vocabulary – labels for parts of a cell.
		Specific detail and elements	Major facts about X, significant events for X.
Conceptual Knowledge	The interrelationships among the basic elements within a larger structure that enable them to function together.	Classification and categories	Different periods of geological time.
		Principles and generalizations	Fundamental laws of physics, major principles of chemistry relevant to life processes.
		Theories, models, and structures	Formulation of the theory of evolution.
Procedural Knowledge	How to do something; methods of inquiry, and criteria for using skills, algorithms, techniques and methods	Subject-specific skills and algorithms	Algorithm for solving X, skills involved in X.
		Subject-specific techniques and methods	Scientific method, methods for evaluating X, techniques used to solve X.
		Criteria for determining when to use appropriate procedures	Criteria for determining which of several types of method X to use.
Meta cognitive Knowledge	Knowledge about cognition in general as well as awareness and knowledge of one's own cognition. Students use this knowledge to appropriately adapt the ways in which they think or operate.	Strategy	Knowledge of rehearsal of information is one way to retain information, planning strategies such as setting goals for reading or self-testing.
		Cognitive tasks, including appropriate contextual and conditional knowledge	Knowledge of the cognitive demands of different tasks, e.g. recall tasks make more demands on memory system than recognition tasks, summarizing results in deeper level of comprehension.
		Self-knowledge	Awareness of one's own knowledge level, strengths and weaknesses in a task.